Elements of AIML
Unit 04 – Lecture 03 Notes
Clustering, Reinforcement Learning & Semi-Supervised ML

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1 Lecture Overview

This lecture covers three important ML paradigms beyond basic supervised learning:

- **Clustering** (unsupervised),
- **Reinforcement learning** (interaction + reward),
- **Semi-supervised learning** (limited labels + many unlabeled examples).

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning* (2e), selected chapters.;Scikit-learn documentation, accessed 2026-02-28.

2 How to use these notes (self-study)

This lecture has three mini-topics. Study them in this order: (1) clustering (because it uses only data X), (2) semi-supervised learning (because it still starts from labeled examples), and (3) reinforcement learning (because it changes the setting from datasets to interaction over time). For each topic, focus on the **objective** and the **main pitfalls**.

3 Learning Outcomes

- Explain clustering and describe k-means steps and objective.
- Identify practical issues in clustering (choice of k, initialization, scaling).
- Explain key RL terminology and the RL learning goal.
- Describe semi-supervised learning and risks of pseudo-labeling.

4 Keywords

- **Cluster:** group of similar points.
- **k-means:** clustering algorithm minimizing within-cluster squared distances.
- **State/action/reward/policy:** core RL terms.
- **Semi-supervised:** learning using both labeled and unlabeled data.

5 Abbreviations and symbols

- k: number of clusters; μ_j: centroid of cluster j; $c(i)$: cluster assignment of point x_i.
- Inertia/objective: $\sum_i \|x_i - \mu_{c(i)}\|^2$ (within-cluster sum of squares).
- RL: s state, a action, r reward, π policy, γ discount factor.

- $Q(s, a)$: action-value (Q-function); α: learning rate.
- Semi-supervised: labeled set L, unlabeled set U, pseudo-labels.

6 Three paradigms at a glance

- **Clustering:** no labels, no rewards. The question is "what groups already exist in the data?".
- **Semi-supervised learning:** a few labels plus many unlabeled examples. The question is "how can unlabeled data help a supervised model?".
- **Reinforcement learning:** no fixed label per row; instead the agent receives delayed rewards while interacting with an environment.

This comparison matters because the three paradigms use different feedback signals and therefore require different evaluation habits.

7 Clustering

7.1 What is clustering?

Clustering is an unsupervised task: we group data points to discover structure. Unlike classification, there are no labels provided.

7.2 Use-cases

- customer segmentation,
- grouping similar documents/images,
- anomaly detection (points far from clusters).

8 k-means

8.1 Algorithm

k-means alternates between assignment and update:

1. initialize k centers,
2. assign each point to the nearest center,
3. recompute each center as mean of assigned points,
4. repeat until assignments stabilize.

8.2 Objective

k-means minimizes within-cluster sum of squares:

$$\sum_{i=1}^{n} \left\| x_i - \mu_{c(i)} \right\|^2$$

8.3 Why the objective makes sense

The objective encourages:

- **compactness:** points in a cluster should be close to the centroid,
- **separation:** different clusters represent different regions of the feature space.

Because it uses Euclidean distance, k-means works best when clusters are roughly spherical and similar in size.

8.4 Practical notes

- scaling matters (distance-based),
- initialization affects result; multiple runs help,
- choosing k is not automatic (elbow method is a heuristic).

8.5 Choosing k and evaluating clustering

Since clustering has no ground-truth labels, we use heuristics and domain checks:

- **Elbow method:** plot inertia vs k and look for diminishing returns.
- **Silhouette score (intuition):** compares how close a point is to its own cluster vs other clusters.
- **Stability:** run k-means multiple times; stable clusters increase confidence.
- **Domain interpretability:** clusters should map to meaningful groups (e.g., customer segments).

8.6 Mini worked example (1D)

Points: $[1, 2, 8, 9]$ with $k = 2$. Initialize $\mu_1 = 2$, $\mu_2 = 9$.

- Assign: $\{1, 2\} \to \mu_1$ and $\{8, 9\} \to \mu_2$.
- Update: $\mu_1 = \text{mean}(1, 2) = 1.5$, $\mu_2 = \text{mean}(8, 9) = 8.5$.

Next iteration keeps the same assignments, so the algorithm converges.

9 Semi-supervised learning

9.1 Motivation

Often labels are expensive (medical, legal), but unlabeled data is abundant.

9.2 Self-training (one simple approach)

1. train a model on labeled data,
2. predict labels for unlabeled data,
3. add high-confidence pseudo-labels to training,
4. repeat.

Risk: if pseudo-labels are wrong, errors can be reinforced.

9.3 Pseudo-labeling failure example

Imagine a medical-image dataset with very few positive cases. If the first model is already biased toward the majority class, it may assign high-confidence negative pseudo-labels to borderline positive cases. Those wrong pseudo-labels are then fed back into training, making the model even less sensitive to rare positives. So semi-supervised learning must be validated carefully; "more data" is not automatically "better data".

9.4 Other semi-supervised approaches (brief)

- **Co-training**: two models teach each other using different feature "views" (when applicable).
- **Label propagation**: build a similarity graph and spread labels to nearby points.
- **Consistency regularization**: encourage stable predictions under perturbations (modern deep learning idea).

10 Reinforcement learning (overview)

10.1 Core idea

RL learns by interaction. At each time step:

- agent observes state s,
- chooses action a,
- receives reward r and new state s'.

Goal: learn a policy $\pi(a|s)$ that maximizes cumulative reward. This is where the agent concepts from **Unit 01 Lecture 02** become directly useful again.

10.2 Return and discounting

We usually care about the **return** (total future reward), not just immediate reward:

$$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots$$

where $\gamma \in [0, 1)$ is the **discount factor**.

- γ close to 0: focus on short-term reward.
- γ close to 1: focus on long-term reward.

10.3 Q-learning (numeric intuition)

One common RL idea is to update the value of an action using both the immediate reward and the estimated future value. Suppose:

- old estimate $Q(s, a) = 2.0$,
- immediate reward $r = 1$,
- best next-state value $\max_{a'} Q(s', a') = 3.0$,

- learning rate $\alpha = 0.5$ and discount $\gamma = 0.9$.

Then the target is:

$$r + \gamma \max_{a'} Q(s', a') = 1 + 0.9 \times 3 = 3.7$$

and the updated value becomes:

$$Q_{\text{new}}(s, a) = 2.0 + 0.5(3.7 - 2.0) = 2.85$$

Interpretation: the old estimate moves partway toward a better estimate based on reward plus promising future outcomes.

10.4 Why RL is harder

- feedback can be delayed,
- exploration vs exploitation trade-off,
- environment can be stochastic.

11 Practice (with answers)

P1. Why does scaling matter for k-means?

Answer: k-means is distance-based. If one feature has a large numeric scale, it dominates the distance calculation and therefore dominates clustering.

P2. Give one RL example with state, action, and reward.

Answer: robot navigation: state = robot position; action = move; reward = +1 for reaching goal, negative for hitting obstacles, small negative per step.

P3. What is the main risk of self-training?

Answer: the model can add incorrect pseudo-labels and reinforce its own mistakes, reducing performance.

12 Common pitfalls (and how to avoid them)

- **Using k-means on unscaled data:** distance-based methods are dominated by large-scale features.
- **Assuming k-means finds the global optimum:** it can converge to local minima; use multiple initializations.
- **Confusing clustering with classification:** clusters are discovered structure, not ground-truth labels.
- **Ignoring exploration in RL:** without exploration the agent may never discover better actions.
- **Self-training error reinforcement:** pseudo-labeling can amplify early mistakes; validate carefully.

13 Exit questions (with answers)

Q1. What is the k-means objective in words?

Answer: minimize the total squared distance between points and their assigned cluster centers.

Q2. Why can k-means produce different results on different runs?

Answer: because random initialization can lead to different local minima; multiple initializations reduce this issue.

Q3. Define state, action, reward, policy.

Answer: state = situation description; action = choice made by agent; reward = feedback signal; policy = mapping from states to actions (or action probabilities).

Q4. One benefit and one risk of semi-supervised learning?

Answer: benefit: uses unlabeled data to improve performance; risk: wrong pseudo-labels can reinforce errors.

14 Summary

Clustering discovers structure without labels, RL learns from rewards, and semi-supervised learning leverages unlabeled data when labels are scarce.

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning* (2e), selected chapters.;Scikit-learn documentation, accessed 2026-02-28.